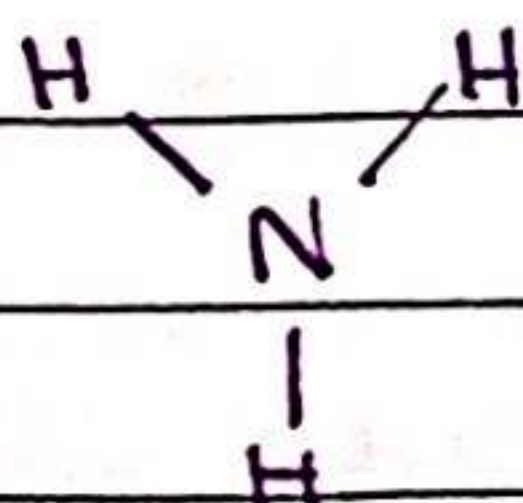


## Molecular formulae

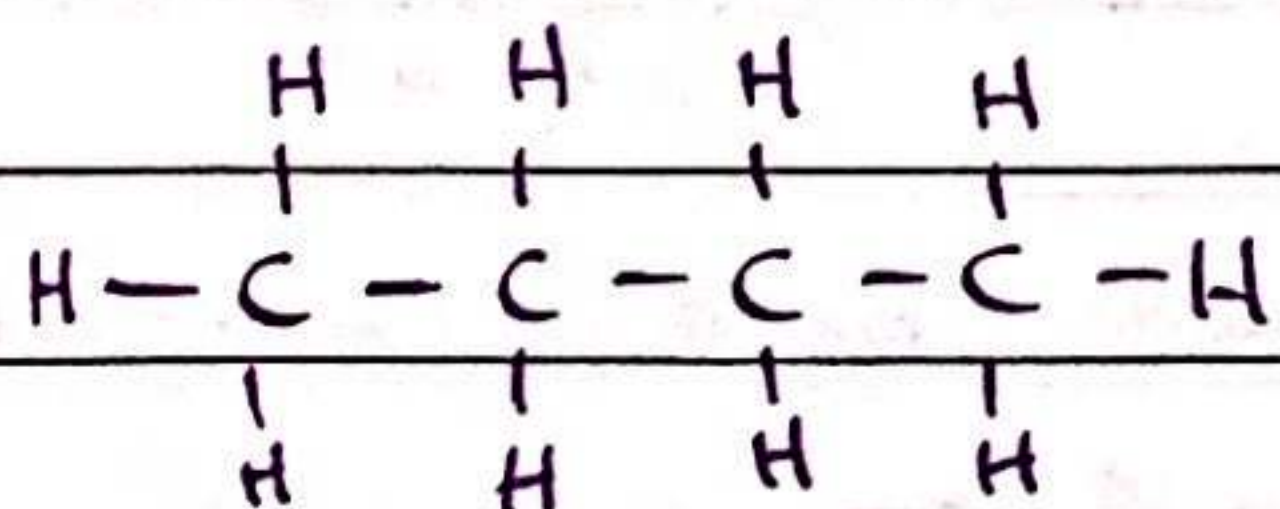
- each element is represented by its own unique symbol. e.g 'H' for hydrogen or 'Na' for sodium
- different compounds have different ratio's of atoms. e.g 'H<sub>2</sub>O' means 2 hydrogen and 1 oxygen
- the chemical formula can be deduced from the relative number of atoms in that compound. e.g 3 atoms of hydrogen + 1 atom of nitrogen = NH<sub>3</sub>
- the displayed formula of 'NH<sub>3</sub>' is:



\* the molecular formula tells you the number and types of different atoms in one molecule

\* the structural formula tells you the way in which the atoms are bonded.

\* the empirical formula is the simplest whole number ratio of the different atoms in a compound.



displayed formula

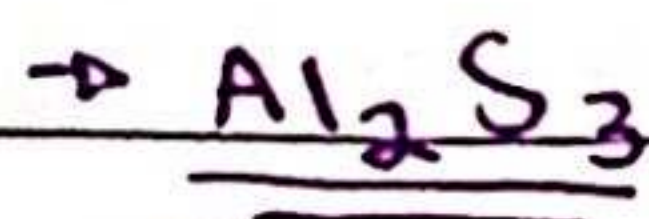
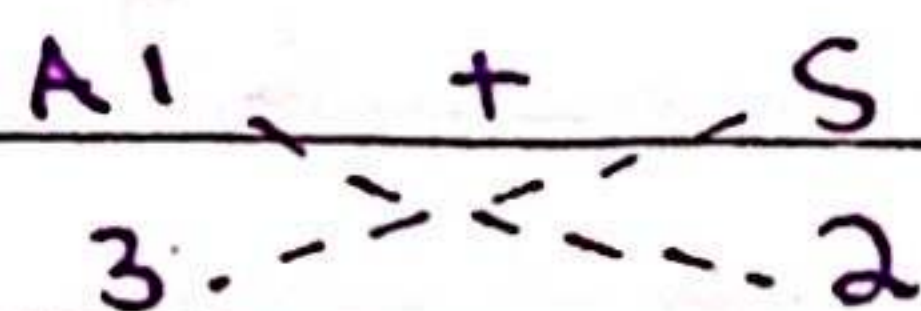
→ butane

CH<sub>3</sub>CH<sub>2</sub>CH<sub>2</sub>CH<sub>3</sub> structural formula

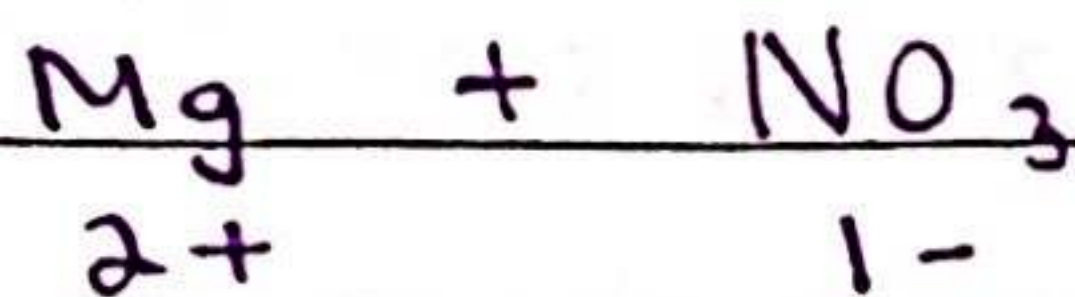
C<sub>4</sub>H<sub>10</sub> molecular formula

C<sub>2</sub>H<sub>5</sub> empirical formula

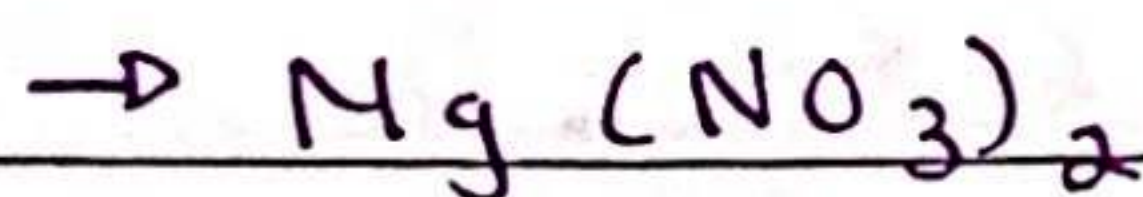
\* we can use the valency of each atom to figure out the formula aluminium sulfide



\* we can use the charge of the ions to figure out the formula of an ionic compound magnesium nitrate



→ two nitrate ions are needed for each magnesium ion



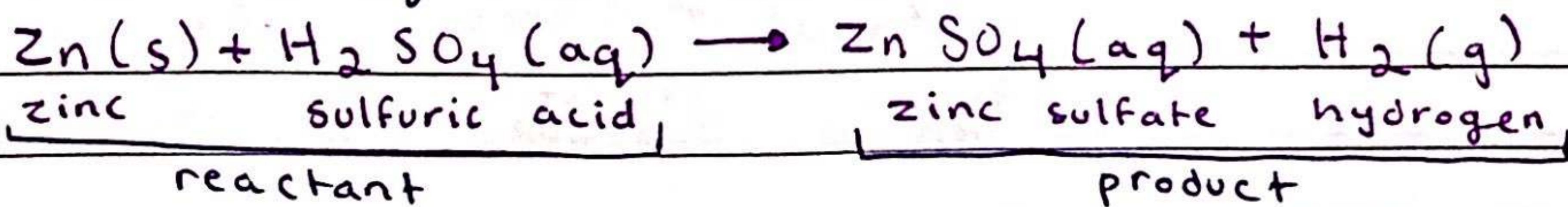
### Equations

word equations :-

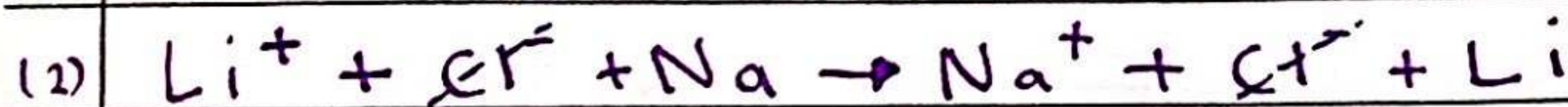
Magnesium + Oxygen → Magnesium oxide

symbolic equations (with state symbols) :-

→ (s) for solid, (l) for liquid, (g) for gas and (aq) for aqueous



ionic equations :-



\* oxidation is loss of electrons

\* reduction is gain of electrons

\* spectator ion is an ion that is not oxidised or reduced during a reaction

### Relative atomic mass ( $A_r$ )

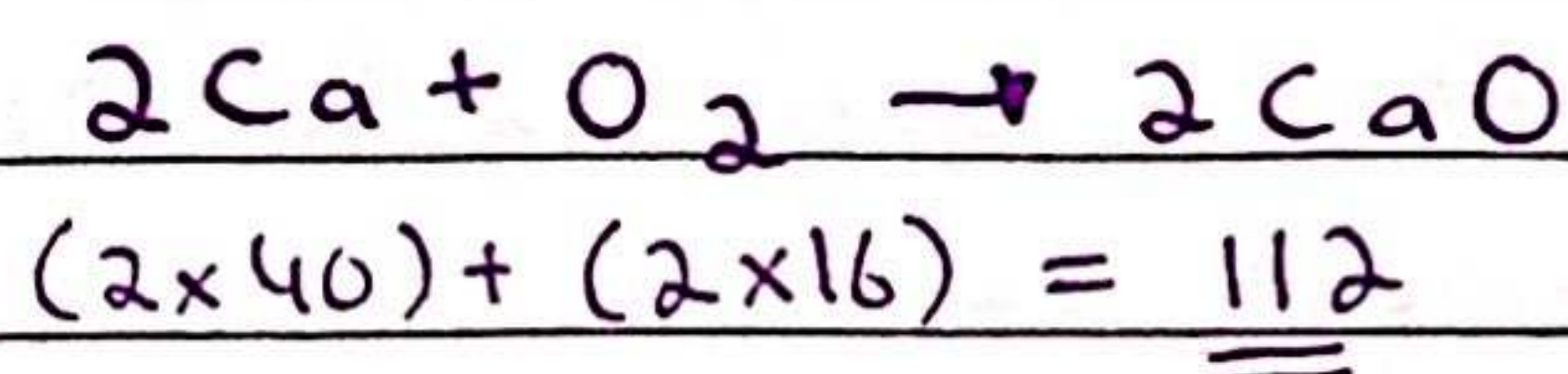
- the average mass of the isotopes of an element compared to  $1/12^{\text{th}}$  of the mass of  $[^{12}_6\text{C}]$

### Relative molecular mass ( $M_r$ )

- the sum of the relative atomic masses

### Law of conservation of mass

- \* mass cannot be created nor destroyed
- \* total mass of reactants = total mass of products



### Mole

- the unit of amount of substance
- one mole =  $6.02 \times 10^{23}$  [atoms, ions, molecules ...]

$$\rightarrow \text{mole} = \text{mass} / M_r$$

$$\rightarrow \text{mole} [\text{gas}] = \text{Volume (r.t.p)} / 24 \text{ dm}^3$$

$$\rightarrow \text{concentration} = \frac{\text{amount of solute}}{\text{volume of solution}}$$

What is the mass of 0.250 moles of zinc?

$$0.250 = \text{mass} / 65$$

$$0.250 \times 65 = \text{mass} \rightarrow \text{mass} = \underline{\underline{16.3 \text{ g}}}$$

How much volume does 3 moles of hydrogen have?

$$3 = \text{volume} / 24$$

$$3 \times 24 = \text{volume} \rightarrow \text{volume} = \underline{\underline{72 \text{ dm}^3}}$$

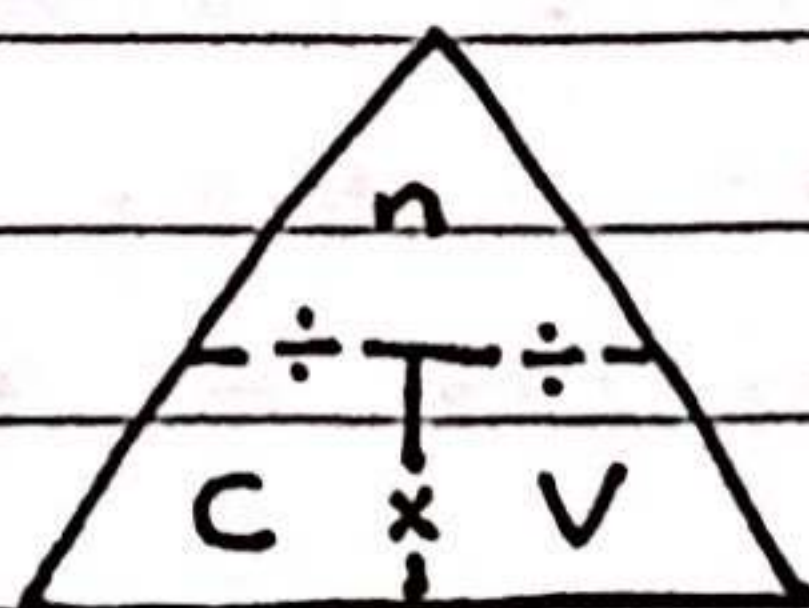
- 10 g of NaOH is dissolved in 2 dm<sup>3</sup> of distilled water. Calculate the concentration (g/dm<sup>3</sup>)

\*  $\text{Concen.} = 10/2$

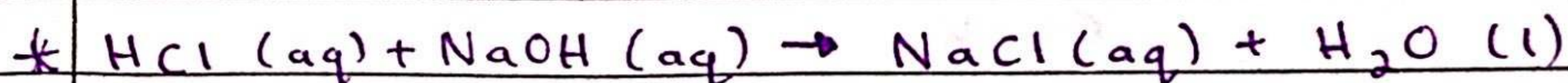
$\text{Concen} = 5 \text{ g/dm}^3$

### Titration

- a method of analysing the concentration of solutions



- 25 cm<sup>3</sup> of HCl is titrated in a solution of 0.1 mol/dm<sup>3</sup> of NaOH and 12.1 cm<sup>3</sup> was required to complete reaction. concentration of the acid?



$\text{mol of NaOH} = (\text{volume} \div 1000) \times \text{Concen.}$

$\text{mol} = 0.0121 \text{ dm}^3 \times 0.1 \text{ mol/dm}^3$

$\text{mol} = 1.21 \times 10^{-3}$

→ acid reacted with alkali with 1:1 ratio

$\text{mol of HCl} = 1.21 \times 10^{-3}$

$\text{concen} = (1.21 \times 10^{-3}) \div 0.025 \text{ dm}^3$

$\text{concen} = 0.0484 \text{ mol/dm}^3$

empirical = ① ÷ the mass of elements by their  $A_r$

② Find molar ratio (÷ by smallest num.)

molecular = ① Find the relative formula mass

② ÷ that by the empirical formula

③ × the number of each element by the number in step 2

### Percentage yield

- yield is the term used to describe the amount of product you can get in a reaction.
- In practice, you can never get 100% yield
  - \* some product could be left in the equipment
  - \* products could be lost during transfer
  - \* side reactions occurring with the air
- Theoretical yield is the amount of product that would be obtained under perfect conditions

$$\rightarrow \text{percentage yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100$$

### Percentage mass

$$\rightarrow \% \text{ of element} = \frac{\text{total mass of the element}}{\text{relative formula mass of compound}} \times 100$$

### Percentage purity

- a pure substance has nothing else mixed with it
- often the product you get is contaminated with unwanted substances, catalysts or other impurities

$$\rightarrow \% \text{ purity} = \frac{\text{mass of pure substance}}{\text{total mass of substance}} \times 100$$